

Severity Agent Walkthrough

Overview

The Severity Agent has been successfully developed and integrated into the MVP. It acts as an intermediate evaluator, interpreting the output of the Disease Detection Agent to provide a structured, user-friendly severity assessment.

Changes Made

- **Severity Agent Module:**

- Located at [severity_agent.py] (file:///d:/agricare/agents/severity_agent/severity_agent.py).
- Uses a deterministic rule-based logic model (rule_based_v1) to assign severity scores based on model confidence.
- Implements the future-proof interface analyze_severity(image_path, disease_class, confidence) .
- Output schema strictly adheres to:

```
{
  "severity": "Medium",
  "severity_score": 2,
  "details": "Moderate confidence indicates intermediate symptom presenta
  "assessment_method": "rule_based_v1"
}
```

- **Healthy Class Override:** Explicitly bypasses the severity logic if the class name contains "healthy" , enforcing a Severity of "None" and a score of 0 .
- **Streamlit Integration:**
 - [1_Disease_Detection.py] (file:///d:/agricare/app/pages/1_Disease_Detection.py) updated to evaluate severity immediately following detection.
 - The disease report now dynamically renders the *Severity Level* and *Assessment Method* in place of the old placeholder.
- **Unit Testing:**

- Comprehensive tests added in [test_severity_agent.py] (file:///d:/agricare/tests/test_severity_agent.py).

Validation Results

- Ran `venv\Scripts\python -m unittest tests/test_severity_agent.py` .
- Ran 4 tests in 0.000s OK
- Passed checks:
 1. `test_healthy_class` : Returned "None"
 2. `test_high_confidence` (95.0%): Returned "High"
 3. `test_medium_confidence` (80.0%): Returned "Medium"
 4. `test_low_confidence` (50.0%): Returned "Low"

The agent is fully functional and ready to pass data to the Advisory Agent!